MECH 325 - Midterm Book 1

Tuesday, October 25, 2016

Name: Solution

Student Number: _____

Circle your section

Section 101:	**T1A (11:00am tutorial)**	**T1B (3:30pm tutorial)**	**T1E (9:00am tutorial)**
Section 102:	**T1C (2:00pm tutorial)**	**T1D (5:00pm tutorial)**	**T1F (9:30am tutorial)**

Signature: _____

Part 1 MC Mark _____ / 14

Part 1 SA Mark _____ / 16

Part 2 LA Mark _____ / 30

Total _____ **/ 60**

Instructions

There are two parts to this exam with different instructions for each. Please read carefully.

Part 1 – Closed-Book (30 marks)

Multiple choice (14 marks: 1 marks per question). Complete all 14 questions by marking your response **in pencil in the computer score card.** Write your name and student number on the computer card and mark your student number in the "ID Field".

Short answer (16 marks: 2 or 4 marks per question). Complete all 6 questions by marking your response in this exam booklet.

After you have completed Part 1, hand in your booklet and Scantron scorecard and you will receive a handout for Part 2 and a separate exam booklet for your answers. You may not return to Part 1 after you hand it in.

Part 2 – Open-Book (30 marks)

Long answer (30 marks): Complete all parts to the questions for Part 2 by marking your responses in the separate exam booklet. This portion of the exam is open-book and open-notes.

Only non-programmable calculators may be used (or programmable calculators with purged memories).

Part 1A: Multiple Choice

Answer all 14 questions. Each is worth 1 mark. Choose the single best response – a given question may have more than one choice that is correct but marks will only be given for the *best* answer.

1. **What is the best description of the type of motion between the teeth of two properly meshed spur gears?**

   a) Purely rolling
   b) Some sliding when teeth engage and disengage, but predominantly rolling
   c) Approximately equal parts sliding and rolling
   d) Some rolling when teeth engage and disengage, but predominantly sliding
   e) Purely sliding

2. **When creating involute profile gear teeth, which of the following diameters is used?**

   a) Pitch Circle
   b) Addendum Circle
   c) Dedendum Circle
   d) Clearance Circle
   e) Base Circle

3. **Which of the following gears allows the shafts to be offset and non-intersecting?**

   a) Spur gear
   b) Planetary geartrain
   c) Helical gear
   d) Bevel gear
   e) Hypoid Gear

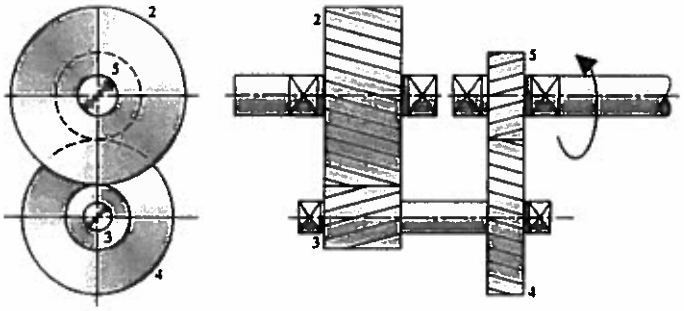
4. **A gear train has gears with pitch diameters of 20 cm (pinion) and 40 cm (gear). The pinion is turned at a speed of n = 300 rev/min by a motor supplying 500 W of power. What is the torque delivered to the gear?**

   a) 5.6 Nm
   b) 23.9 Nm
   c) 25.0 Nm
   d) 31.8 Nm Typo in eqn
   e) 53.7 Nm

5. **A spur gear with a pressure angle of 14.5° has 17 teeth and a pitch diameter of 2.5 in. The gear transmits 1 hp at 1800 rpm. If the gear is replaced by a spur gear with 24° pressure angle, with all other quantities remaining the same, which one of the following will increase?**

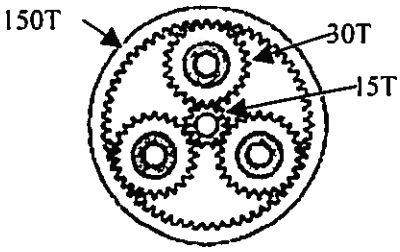
   a) Diametral pitch
   b) Torque transmitted
   c) Transmitted force
   d) Radial force
   e) Axial force

6. **On the following gear set, gear 5 is the driving pinion. Gear 5 has 19 teeth, Gear 4 has 38 teeth, Gear 3 has 16 teeth and gear 2 has 64 teeth. A 1600 rpm motor is driving the gear box in the direction shown. What are the speed and direction of the output shaft, relative to the drawing on the left?**



a) 300 rpm clockwise
b) 300 rpm counter-clockwise
c) 200 rpm clockwise
(d) 200 rpm counter-clockwise
e) 600 rpm clockwise

7. **For the planetary gear train shown, the ring gear rotates at 100 rpm counter-clockwise and the arm is fixed. How fast does the sun gear rotate and in which direction?**



a) 500 rev/min (counter-clockwise)
b) 500 rev/min (clockwise)
c) 1000 rev/min (counter-clockwise)
(d) 1,000 rev/min (clockwise)
e) 1,500 rev/min (counter-clockwise)

8. **A power screw engages with a nut with 10 threads. Approximately how much of the load is carried by the first 7 engaged threads, combined, in the power screw?**

a) 10%
b) 20%
c) 50%
d) 80%
(e) 100%

9. **Rank the following flexible drives from lowest to highest power capacity. Assume approximately equal dimensions and operating conditions.**

a)	V-Belt	Flat belt	Roller Chain
b)	V-Belt	Roller Chain	Flat belt
(c)	Flat belt	V-Belt	Roller Chain
d)	Flat belt	Roller Chain	V-Belt
e)	Roller Chain	Flat belt	V-Belt

10. **A V-belt drive has a driving pulley operating at 1800 rpm, a drivetrain value of e = 0.33, a nominal power of H_{nom} = 0.75hp, a service factor of K_s = 2.2, and a design factor of n_d = 1.5. TWO belts are to be used in parallel. The following belts are available in the catalogue. Which is the smallest belt that will satisfy the design requirements?**

a) Allowable power H_a = 1.0 hp

b) Allowable power H_a = 1.4 hp

c) Allowable power H_a = 1.5 hp

d) Allowable power H_a = 2.5 hp

e) Allowable power H_a = 5.0 hp

11. **Which of the following is the best description of how a hydrodynamic bearing works?**

a) The shaft moves by sliding against the bearing material

b) High pressure fluid is injected into the bearing to support the shaft

c) The rotating shaft pulls fluid to form a layer between it and the bearing

d) The shaft is supported by high pressure air fed through a porous material

e) Metal-on-metal contact is avoided by packing the bearing with a thick grease

12. **Which of the following design or operating parameters should be increased in order to be able to reduce the required bushing length of a boundary-lubricated bearing in service? (assume all other parameters remain fixed)**

a) Operating rotation speed, N

b) Applied force, F

c) Coefficient of friction, f,

d) (PV) rating of the bushing

e) Wear factor, K

13. **Which of the following is NOT a possible or recommended consideration when significantly increasing the power carried by a roller chain drive? (assume the linear speed is moderate)**

a) Increasing chain pitch size

b) Upgrading the lubrication system

c) Increasing both driving and driven sprocket diameters

d) Increasing the number of teeth on the driving sprocket only

e) Using a higher strand chain (e.g., duplex or triplex instead of simplex)

14. **What type of bearing is depicted in the image to the right?**

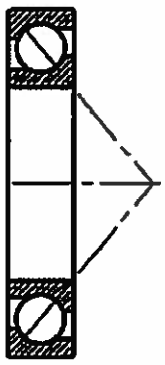
a) Deep-groove ball bearing

b) Angular contact bearing

c) Tapered roller bearing

d) Tapered ball bearing

e) Self-aligning bearing



Note: Total out of 13.

Part 1B – Short-Answer Questions (16 marks)

Answer all questions in the spaces provided. Each question is worth 2 marks.

15. a) In gear design, what is meant by interference? How can it be avoided (2 marks)?

Interference relates to gear tooth contact that is not conjugate. It causes gear teeth to bind and cause excessive wear.

Interference is avoided by having the gear teeth engage between the tangents on the base circles formed by the pressure angle. One can select gears with greater number of teeth.

1 mk

1 mk

b) Two parallel shafts with 6 inch center distances are to be connected by a pair of gears having a diametrical pitch of 16 teeth/inch. The spur gears have a contact pressure angle of 24° and provide a gear reduction of 3.0. Determine the pitch diameter and number of teeth in the pinion gear (2 marks).

$r_p + r_G = 6''$ $P = 16$ teeth/inch $e = 3:1 = 1/3$

so $r_G/r_p = 3$

$\therefore 4r_p = 6''$ $r_p = 1.5''$ and $r_G = 4.5''$

So $d_p = 3''$ and $d_G = 9''$

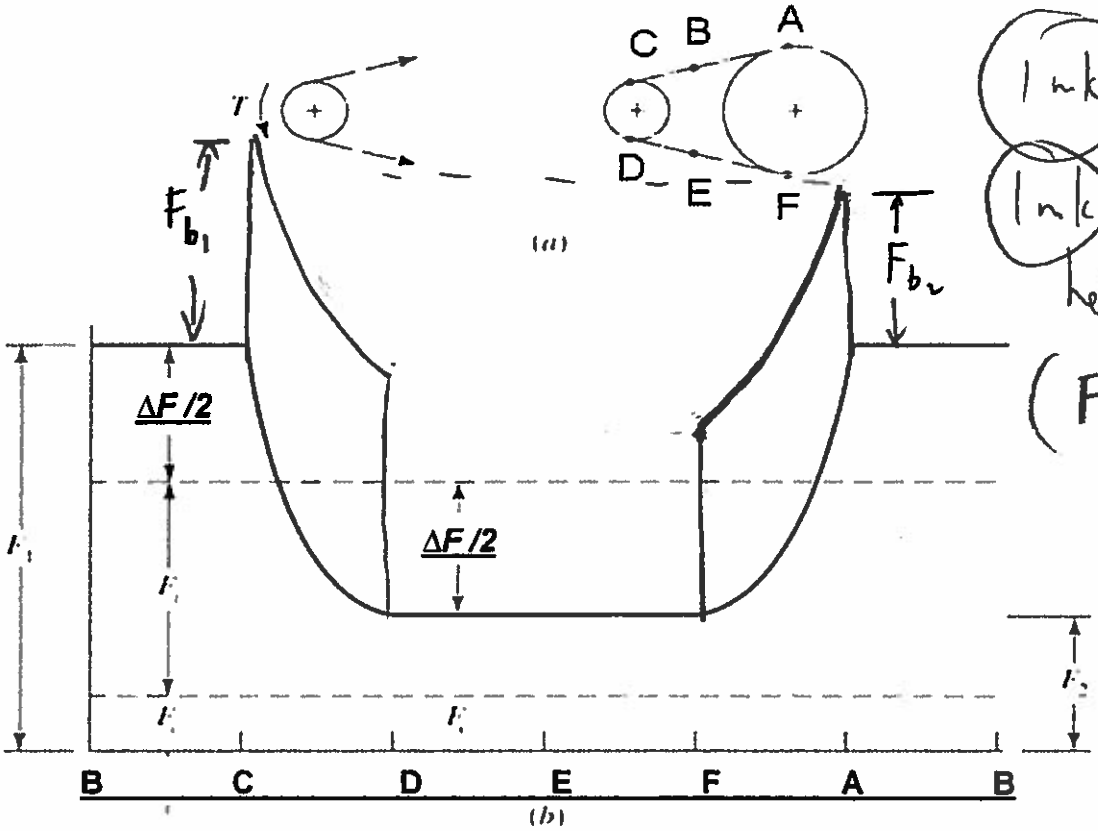
$3'' \times 16 = 48$ teeth

$9 \times 16 = 144$ teeth

1 mk right idea

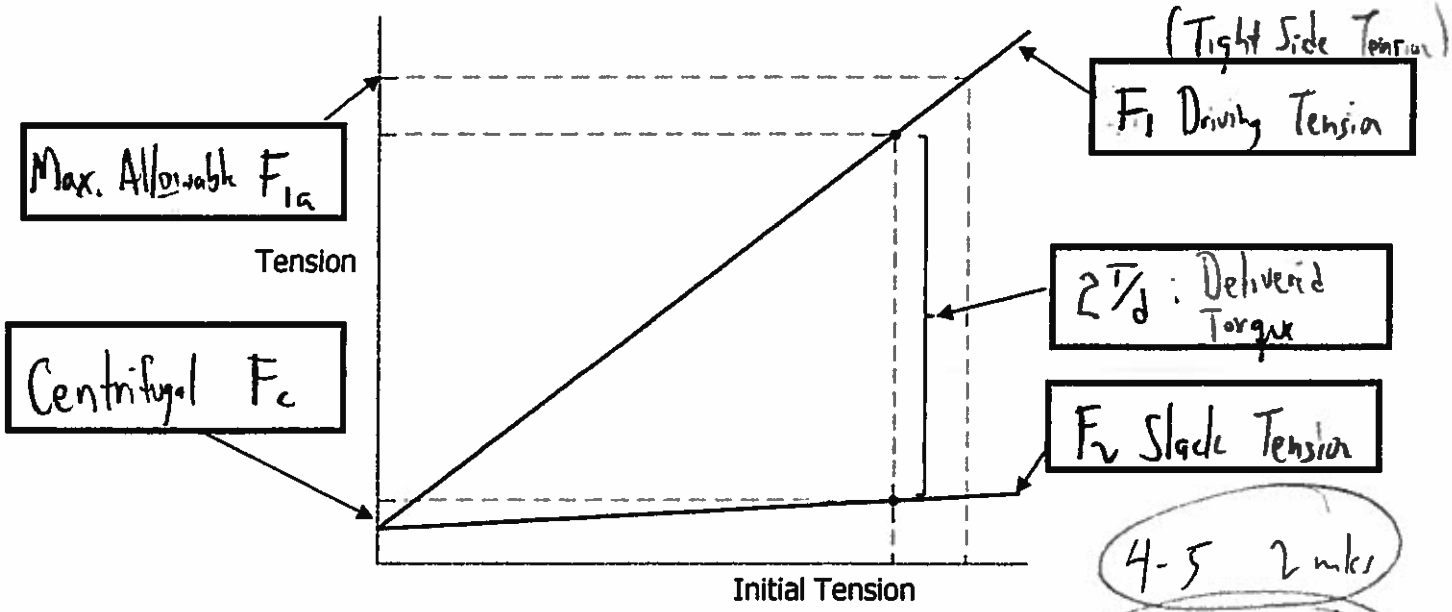
1 mk correct solution

16. The diagram shown below is the tension profile for a flat belt. On this diagram, sketch the tension profile for a V-belt belt that is transmitting the same torque, but otherwise has the same linear speed, initial tension, pulley diameters, and belt weight per unit length. (2 marks)



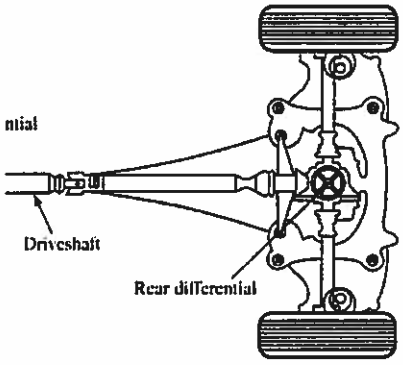
1 mk for sketch
1 mk for differing heights
($F_{b_1} > F_{b_2}$)

17. The relationship for flat belt tension versus initial tension is shown in the diagram below. Complete the diagram by giving the name or equation for each of the four quantities identified. Put your answers in the boxes provided (2 marks)



4-5 2 mks
1-3 1 mk
0 0 mk

18. The following figure illustrates a rear differential of a rear wheel drive vehicle.



a) Explain why a differential gearbox is used on this vehicle. (2 mark)

A differential is used in order that the car can turn in a circle. It provides torque to each wheel while compensating for the different speeds due to varying turning arc.

1 mk

1 mk

b) If you happen to park the car with one of the wheels on ice and the other on dry pavement, will you be ~~able to~~ leave the parking spot when it is time to leave? Explain. (2 mark)

1 mk

No. No torque can be transferred to the wheel on dry pavement. Hence, the wheel on ice will spin while the other remain stationary.

1 mk

19. A power screw is self-locking when the screw will not turn once a lifting torque has been removed. The equations for the raising torque, T_R, and the lowering torque, T_L for a power screw are given below:

$$T_R = \frac{Fd_m}{2}\left(\frac{\pi f d_m + l}{\pi d_m - f l}\right) \quad T_L = \frac{Fd_m}{2}\left(\frac{\pi f d_m - l}{\pi d_m + f l}\right)$$

What is the condition for self-locking? Derive the thread efficiency equation for the power screw for the raising condition. (2 marks)

Self locking when $\pi f d_m - l > 0$

$\therefore \pi f d_m > l$

$f > l/\pi d_m$ (1 mk)

Thread eff $= \eta = \frac{T_0 (\text{no friction})}{T_R (\text{with friction}}$ with $f = 0$ $T_0 = \frac{Fl}{2\pi}$

$\therefore \eta = \frac{Fl}{2\pi T_R}$ (1 mk)

20. A) Why is lubrication important in a gear train system? (1 mark)

B) What is meant by "galling?" (1 mark)

A)
- avoid metal to metal contact
- reduce friction
- provide cooling

(1 mk)

B) Some metals (titanium, aluminum) will form a 'welded' bond when subject to contact under stress. These bonds will undergo shear and fail causing metal erosion.

(1 mk)

Long Answer Problem (30 marks)

By attaching an idler gear between the pinion and gear, one can create a 3-gear geartrain where the output shaft turns in the same direction as the input shaft (the pinion meshes with the idler while the idler meshes with the gear). Specifications for the gear box are as follows:

You can work in either Imperial or SI units. However, do not mix units as the conversions are not exact.

Input Power to pinion	20 hp (15kW)
Driving Motor	2 cylinder gasoline engine.
Driven Load	moderate shock
Speed of pinion	2500 rpm
Pinion Diametrical Pitch	6 teeth/inch (module = 4 mm.)
Face width of teeth	2 inches (50 mm.)
Number of Teeth on Pinion	17 teeth
Number of Teeth on Idler Gear	20 teeth
Gear Reduction Ratio	3.5:1
Pressure Angle	20°
Gear Material (all 3 gears)	Grade 2 Steel
Brinell Hardness Pinion (through hardened)	325 Bhn
Brunel Hardness Idler (through hardening)	325 Bhn
Brinell Hardness Gear (through hardened)	275 Bhn
Gear Quality (all 3 gears)	6
Design Lifetime (all 3 gears)	10^8 cycles
Reliability (all 3 gears)	0.90
Size Factor, K_s (all 3 gears)	1.0
Load Distribution Factor K_m (all 3 gears)	1.6
Rim Thickness Factor K_B (all 3 gears)	1.0

21. Determine a) the number of teeth on the gear, b) the centre line distance from the pinion to the gear, c) the force transmitted from the pinion to the idler and from the idler to the gear, and d) the radial load on the pinion shaft and the idler shaft. **(9 marks).**

22. Assuming that the transmitted load on the pinion gear is 450 lbf (2000N), determine all the AGMA factors for the gear tooth bending stress equation for the pinion, and the idler. **Estimate the bending stress number for the pinion and idler gears.** Justify all assumptions. **(10 marks)**

23. Assume that the gear bending stress number for the pinion is 10000 psi (70MPa), and the idler is 13000 psi (90Mpa). **Estimate the minimum safety factor for bending stresses for the pinion and idler gears. (8 marks)**

24. This geartrain is to operate 8 hours a day during regular business days (Monday through Friday). Given the calculations you have performed, are you confident the geartrain will operate for at least five years? Explain. Which gear would you expect to fail first? Why? Are there any recommendations you can make to the client to improve the lifetime of the gearbox? **(3 marks).**

Problem 21

Pitch = 6 teeth/inch module = 4 mm.

9 mk total

a) Number of teeth on gear (1 mk)

$N_g = e\,N_p = 3.5(17) = 59.5 = \underline{59 \text{ or } 60 \text{ teeth (both acceptable)}}$

b) dia. of pinion = $N_p/\text{pitch} = 17/6 = 2.83$ inches | dia. of pinion = $N_p m = 17 \times 4\text{mm} = 68\text{mm}$.

dia. of idler = $20/6 = 3.33$ in. | dia of idler = $20 \times 4 = 80$ mm

dia. of gear = $60/6 = 10$ in. | dia of gear = $60 \times 4 = 240$ mm

or (dia of gear $59/6 = 9.83$ in) | (dia of gear $= 59 \times 4 = 236$ mm

(1 mk formulation)

∴ Center line distance = $r_{pinion} + d_{idler} + r_{gear}$

Center line distance = $\frac{1}{2}(2.83'') + 3.33'' + \frac{1}{2}(10'')$ | Center line distance = $\frac{1}{2}(68) + 80 + \frac{1}{2}(240\text{mm})$

$= \underline{9.75 \text{ inches}}$ or $\underline{9.66 \text{ in}}$ (1 mk solution) | $= \underline{234 \text{ mm.}}$ or $\underline{232 \text{ mm}}$

c) Power = Torque × speed

∴ T_{pinion} = power/speed $= \dfrac{(20 \text{ h.p.})(550 \text{ ft·lb/s})(12 \text{ in/ft})}{(2500 \text{ rev/min})(1 \text{ min}/60 \text{ sec}) \times 2\pi}$

$T_{pinion} = 504 \text{ lbf·in}$

(2 mk formulation)

Torque = Force × distance

∴ $T_{pinion} = W_t \times (d_{pinion}/2)$

$W_t = T/d_2 = 504/(2.83''/2)$

$\underline{W_t = 356.2 \text{ lbf}}$ [May also use $W_T = 33000\,H/V$]

Note: Some students will include total force as transmitted load. This is acceptable i.e. $F_t = 379$ lbf

Metric: $W_t = \dfrac{60000\,H}{\pi d n} = \dfrac{60000(15\text{kW})}{\pi(68\text{mm})(2500\text{rpm})}$ (1 mk)

$\underline{W_t = 1.69 \text{ kN}}$

Same force is translated from idler to gear (1 mk)

∴ $\underline{W_{tp} = W_{tg} = 356.2 \text{ lbf (or } 1.69 \text{ kN)}}$

d) i) Radial load on pinion



$W_R = W_t \tan\phi$ (1 mk)

$= (356.2)(\tan 20°)$

$\underline{W_R = 129.6 \text{ lbf}}$

Metric: $W_R = (1.69\text{kN})(\tan 20) = 0.615\text{kN}$

ii) Radial load on idler = $\underline{0}$ (equal and opposite forces) (1 mk)

P.22 $W_t = 4501bf$

10mk Total

$$\sigma = W^t K_o K_v K_s \frac{P_d}{F} \frac{K_m K_B}{J}$$

AGMA bending strength number equation

Given $K_m = 1.6$ (all gears)

$K_s = 1.0$ (all gears)

K_o = overload factor — driven load = moderate shock

driving load = 2 cyl. gas engin = light shock

$\therefore K_o = 1.5$ (all gears) (2mk) Note: 1 mk for $K_o = 1.25$!

K_v = dynamic factor (Note: $K_v = \left(\frac{A+\sqrt{V}}{A}\right)^B$ $B = 0.25(12-Q_v) = 0.8255$ ($Q=6$) $A = 50+56(1-B) = 59.77$

pitch line velocity $V_t = \omega r = (2500 \text{ rev/min})(2\pi)\left(\frac{2.83''}{2}\right)\left(\frac{1 ft}{12 in}\right)$

$= 1852$ ft/min (1 mk)

From Figure 14-9 @ $Q_v = 6$ $K_v \simeq 1.6$ (all gears)

(Note: $K_v = 1.56 \sim 1.6$ for $K_v = \left(\frac{A+\sqrt{V}}{A}\right)^B$) (2mk)

$K_B = 1.0$ all gears are solid

Bending Strength Geometry Factor

* assume single tooth loading (worst case)

i) pinion into idler 17 into 20

$J = 0.30$ (Figure 14-6) (1 mk)

ii) idler into pinion 20 into 17

$J = 0.31$

iii) idler into gear 20 into 60

} both conditions should be checked (1mk)

$J = 0.33$

$J = 0.31$ min.

P. 22 cont.

ii) $\sigma = W_T K_o K_v K_s \frac{P_d}{F} \frac{K_m K_B}{J}$

$\sigma = (450 lb_f)(1.5)(1.6)(1.0) \frac{6 \text{ teeth/inch}}{2 \text{ inches}} \frac{(1.6)(1.0)}{J}$

$\sigma = \frac{5184}{J}$ psi

formulation 1 mk

i) pinion $J = 0.30$

$\sigma = 5184/0.30 = \underline{17280 \text{ psi}}$ 1 mk

ii) idler $J = 0.31$ or 0.33. Choose lower value $\therefore J = 0.31$

$\sigma = 5184/0.31 = \underline{16722 \text{ psi}}$ 1 mk

Problem 23.

8 mk total

$\sigma_{all} = \frac{S_t}{S.F} \frac{Y_n}{K_T K_R}$ S.F. = safety factor; S_t = allowable bending stress number

From Fig. 14-2 @ $H_B = 330$ Bhn

$S_{t\,pinion} = S_{t\,idler} \doteq 49\text{-}51$ kpsi Say $S_t = 50$ kpsi

or From curve fit equation (Fig. 14-2) $S_T = 102 H_B + 16400$ psi

$S_{t\,pinion} = S_{t\,idler} = 102(330) + 16400$ 2 mk

$S_t = 50$ kpsi [345 MPa]

Stress Cycle Factor:

Use Fig 14-14 @ cycles $= 10^8$ $Y_n \approx 0.95$ 1 mk

Temp. $K_T = 1.0$ *assume $T < 250°F$* 1 mk [state assumption]

Reliability $K_R = 0.85$ Table 14-10 @ $R = 0.90$ 1 mk

Now $(S.F)(\sigma_{working}) = S_t \frac{Y_n}{K_T K_R}$ 1 mk formulation

P. 23 cont.

i) Pinion: $S_{.}F(10{,}000\ psi) = 50000\ \frac{0.95}{(1.0)(0.85)}$ (1 mk)

$\doteq 56{,}000\ psi$

$\therefore \underline{S_{.}F = 5.6}$ (range 5.5-5.7)

ii) Idler: $S_{.}F(17{,}000) = 56{,}000\ psi$ (1 mk)

$\underline{S_{.}F = 4.3}$ (range 4.2-4.4)

Metric pinion $S_{.}F(70\ MPa) = (354)\frac{0.95}{(1.0)(0.85)}$

$\underline{S_{.}F. = 5.65}$ (range 5.5-5.8)

idler $S_{.}F(90\ MPa) = \frac{354(0.95)}{(1.0)(0.85)}$

$\underline{S_{.}F = 4.4}$ (range 4.3-4.5)

Problem 24. Operate for 5 years without breakdown (3 mk Total)

$(8\ hrs/day)(60\ min/hr)(5\ days/wk)(50\ weeks/year)(5\ years)$

(1 mk) $\times 2500\ rev/min$ (approx 10 holidays/year)

$= 1.5 \times 10^9$ cycles (1.56×10^9 for 52 weeks)

This value exceeds our design lifetime of 10^8 cycles so the values of Y_L are needed to be adjusted downward. The calculations need to be re-done.

(1 mk) We expect the idler to fail first since it is undergoing twice the loading (pinion + gear) so it sees 3.0×10^9 cycles/5 years and its $S_{.}F.$ is the lowest.

Recommendations to improve performance: (1 mk)

1) higher quality gear
2) increase face width
3) larger gears
4) other ideas